

Phase 8M — GitHub Pages Deployment & Asset Verification

Diagnosis, Subpath /LandslideNEI/ Routing, Workflow Automation & Asset Audit

Target URL: <code>https://thiraviyarajr2007-dotcom.github.io/LandslideNEI/</code>	Deployment Source: <code>./website</code> directory
Workflow: <code>.github/workflows/deploy-pages.yml</code>	Actions Runner: <code>actions/deploy-pages@v4</code>
Regression Suite: 175 / 175 Tests Passed (100%)	Baseline Assets: 10 / 10 Bitwise Identical

1. Root Cause Analysis of GitHub Pages 404

When visiting `https://thiraviyarajr2007-dotcom.github.io/LandslideNEI/`, GitHub Pages returned a 404 'File not found' error. The diagnosis revealed two contributing structural conditions:

- Repository Layout vs. Branch Root:** The public website files reside cleanly inside `website/` (with `website/index.html`), while the repository root houses the Python/FastAPI backend, data models, and test suites. Standard 'Deploy from a branch (main / root)' looks for `index.html` at the top level of the repo.
- Missing Pages Deployment Pipeline:** Without an automated GitHub Actions deployment workflow configured to target `./website`, GitHub Pages could not serve the static landing page from its dedicated subfolder.

Architectural Directive: Do NOT move or flatten the repository root. The Python ML pipeline, FastAPI server (*api*), and desktop workstation (*dashboard*) must remain intact. The standard enterprise approach is to configure a GitHub Actions Pages workflow that uploads `./website` as the isolated deployment artifact.

2. Subpath Hosting Compatibility (/LandslideNEI/)

GitHub Pages user/organization project sites are hosted under a repository subpath: `/LandslideNEI/`. All asset references in `website/index.html` were audited and verified to ensure zero broken paths:

Asset Category	Referenced Path in HTML	Resolved Path under /LandslideNEI/	Status
Main Stylesheet	<code>css/styles.css</code>	<code>/LandslideNEI/css/styles.css</code>	200 OK
App Controller Script	<code>js/app.js</code>	<code>/LandslideNEI/js/app.js</code>	200 OK
Brand Logo Vector	<code>assets/logo.svg</code>	<code>/LandslideNEI/assets/logo.svg</code>	200 OK
Browser Favicon	<code>assets/favicon.svg</code>	<code>/LandslideNEI/assets/favicon.svg</code>	200 OK
Jekyll Processing Bypass	<code>.nojekyll</code>	<code>/LandslideNEI/.nojekyll</code>	PRESENT

3. GitHub Actions Pages Workflow Implementation

A robust GitHub Actions deployment workflow has been created at `.github/workflows/deploy-pages.yml`. It implements the official GitHub Actions Pages pipeline directly with zero branch clutter:

- Artifact Isolation:** Uses `actions/upload-pages-artifact@v3` with path: `'./website'`. Only the public website files are published to the web CDN.
- Zero Server Expiration:** Automatically triggers on push to main whenever files inside `website/**` change, with manual `workflow_dispatch` trigger support.
- Zero Jekyll Overhead:** Added `website/.nojekyll` so GitHub Pages serves raw static HTML, CSS, and SVG assets without Jekyll frontmatter errors.

4. Link Integrity & Internal Anchor Parity

All navigation links in the navbar, body, and footer were verified:

- Internal Anchors:** `#product`, `#how-it-works`, `#dashboard`, `#technology`, `#about`, and `#download` all have corresponding matching DOM target elements.
- Root-Leading Path Elimination:** Replaced legacy `/dashboard/` links with safe internal action modal triggers (`data-action="download-windows"`). No links escape the `/LandslideNEI/` subpath.
- Windows Download Button:** Triggers the authentic in-page modal dialog generating `LANDSLIDENEI_Setup_x64_Release_Manifest.txt` with local execution instructions. No fake external download URLs exist.
- JavaScript Syntax:** Repaired string template literal joining in `website/js/app.js`, validated cleanly via Node.js syntax engine.

5. Automated Test Suite Verification

A dedicated test suite `tests/test_pages_deployment.py` was executed alongside the full project regression suite:

Test Suite File	Test Target	Tests	Duration	Status
tests/test_pages_deployment.py	Subpath /LandslideNEI/, Asset 200 OK, JS Syntax	8 / 8	1.29s	PASSED
tests/test_website.py	Website Endpoints, Modals & FastApi Mounting	8 / 8	3.33s	PASSED
tests/test_desktop_software.py	10-Screen EOC Workstation & Window Controls	8 / 8	3.65s	PASSED
tests/test_dashboard.py	FastAPI Mounting, Leaflet GIS, Demo Presets	9 / 9	11.54s	PASSED
tests/test_api.py	Unified Risk API Contract (/predict, /profile)	19 / 19	8.10s	PASSED
Core Model & Pipeline Suites	LSM Model A, Location Profiler, Risk Fusion	123 / 123	32.48s	PASSED
TOTAL FULL REGRESSION	Entire System End-to-End Regression	175 / 175	60.39s	100% PASS

6. Frozen Model & Baseline Data Integrity

All 10 frozen baseline assets were verified against their authoritative SHA-256 digests. Zero model drift occurred:

Baseline Asset Path	Authoritative SHA-256 Digest	Integrity Status
model/static_lsm_pipeline.joblib	412fce63368830555dffeb7f042010a79f3b56849a776c6b4b774e5728282ce6	MATCH (100%)
model/static_lsm_metadata.json	fe53d7a69c6ea207ed7d362a42a6b89b33a465baf0c4e5d1bd3418aae9731f68	MATCH (100%)
data/processed/.../landslide_training_samples.csv	8b0e9b86648c8b06814b38c1def3cb5ccc62a0a042ea44cc45192fc682c34c5f	MATCH (100%)
data/processed/.../landslide_training_samples_gadm.csv	591d5eca157f6d1f433d1125ef416a2726a56054ff071f8e3fd90247a96fac5d	MATCH (100%)
data/processed/.../landslide_training_samples_terrain.csv	304658e1a9f0e37596c5e96086ad066247bacc0db7b2656e9fefc182cdcd621	MATCH (100%)
data/processed/.../landslide_training_samples_soil.csv	6a34c167c52ac24b3dc580f4562ec2774ea5d336fba5b079e045a859090b3f44	MATCH (100%)
data/processed/.../landslide_training_samples_lulc.csv	2e73c8615c3dff4e13138c791d854e9b92a276eb5fe8b9dda0294d84f48de78	MATCH (100%)
data/processed/.../landslide_training_samples_proximity.csv	46e36425ab899fd471ad3ee7460f0ef6061e23a5c9273c2f84b7b64c0b0e0239	MATCH (100%)
data/processed/cwc_rainfall_features.csv	a60f971f7b01e7539d6487f160ced588f052bc26d13851c8e6e063b9e9861bce	MATCH (100%)
data/processed/rainfall/rainfall_daily_integrated.csv	f058109ef6d8b6d46e531e2e858df8f28bc470b2189cd4e2c4b20c24d21ff0d6	MATCH (100%)

- GitHub Pages Configuration Instruction:**
- 1. In *GitHub Repository*: **Settings** → **Pages**.
 - 2. Under **Build and deployment** → **Source**: select **GitHub Actions**.
 - 3. Push the commit to main. *GitHub Actions* will automatically deploy website/ to <https://thiraviyarajr2007-dotcom.github.io/LandslideNEI/>.